

Volatility Forecasting: Diagnostic Appendix

Eddie Liu

A / Verification Summary

Check	Observed evidence
Run coverage	6 neural specifications x 178 folds = 1,068 fits. Nine complete forecast series, each with 3,738 identical dates and targets.
GARCH fitting	228 parameter refits; all convergence flags are zero. 0 / 4778 consecutive forecasts repeat exactly.
Forecast validity	All final forecasts are finite and positive; 0 at or below 1e-12 across all nine series.
Regression suite	Tests cover the actual arch fixed-parameter reference, new-return response, forecast timing, transforms, input validity, sequences, bootstrap mechanics, and CPU training/resumption.
Metric verification	Independent array calculations reconcile with the repository MSE and QLIKE functions. All model CSV hashes match their completion records.

Repaired daily recursion

$$h_t = \text{omega} + \alpha r_{t-1}^2 + \text{beta } h_{t-1}$$

On refit dates, parameters and endpoint variance come from an expanding fit using returns through t-1. Between refits, parameters stay fixed while every new return updates the state. Internal percent-squared variance is divided by 10,000 for the output units.

The corrected experiment does not certify upstream price history against an independent vendor, every unused legacy notebook, or production controls. A converged fit is a numerical condition, not proof that the model is correct.

B / Training and Numerical Checks

Neural model	Mean epochs	Train loss	Validation loss	Paired gap
Pure LSTM	27.1	0.8001	0.8252	+0.0251
Pure GRU	25.5	0.8016	0.8247	+0.0231
Feature LSTM	26.2	0.8009	0.8257	+0.0248
Feature GRU	25.2	0.8023	0.8251	+0.0228
Ratio LSTM	27.1	0.9344	0.9681	+0.0337
Ratio GRU	26.3	0.9396	0.9700	+0.0303

Table A1. Means across 178 folds. Train and validation losses are evaluated in inference mode at the same restored best-validation checkpoint. The losses are standardized-target MSE, not original-scale variance MSE.

Interpretation

The corrected paired gap fixes the old practice of subtracting independent minima from different epochs. A gap is still a diagnostic, not a theorem about overfitting. Pure/feature log targets and ratio targets differ, so cross-family gap magnitudes are not directly comparable.

Transform and reconstruction

For log targets, the forward map uses $\log(\max(y, 1e-8))$; the inverse exponentiates without subtracting epsilon. Clipping is not invertible below its floor, and the tests explicitly check this boundary behavior.

For ratio hybrids, the signed log-ratio target is standardized without taking another logarithm. Reconstruction uses $\exp(\log(g) + \text{predicted } z)$ in float64 and rejects non-finite or nonpositive numerical results.

Reproducibility and interruption handling

Seeds are set before model construction. TensorFlow deterministic CPU operations are enabled. A run signature binds input bytes, relevant code, configuration, and split selection. An output directory with a conflicting signature is rejected. A split is complete only when predictions, required gate rows, and the final training log agree on coverage.

Deterministic smoke tests reproduce within the tested environment. This is not a guarantee of bitwise portability to another TensorFlow version or hardware platform.

C / Paired Loss-Difference Intervals

MSE differences / 10^{-7}

Candidate vs GARCH	Difference	95% lower	95% upper
Lagged RV21	+0.27145	+0.00863	+0.78988
EWMA 0.94	+0.14756	-0.02708	+0.47145
Pure LSTM	+0.33813	-0.04210	+1.08428
Pure GRU	+0.31863	-0.06924	+1.04830
Feature LSTM	+0.36353	-0.05193	+1.17966
Feature GRU	+0.36948	-0.02465	+1.16004
Ratio LSTM	+0.26254	-0.06901	+0.86248
Ratio GRU	+0.25065	-0.06330	+0.81592

QLIKE differences

Candidate vs GARCH	Difference	95% lower	95% upper
Lagged RV21	+0.10836	+0.07105	+0.15051
EWMA 0.94	+0.04126	+0.01967	+0.06587
Pure LSTM	+2.01296	+1.64586	+2.47113
Pure GRU	+2.03095	+1.68416	+2.43884
Feature LSTM	+1.85468	+1.59232	+2.15396
Feature GRU	+1.94634	+1.66693	+2.25586
Ratio LSTM	+1.80176	+1.56706	+2.05614
Ratio GRU	+1.88947	+1.64687	+2.15347

Each interval is pointwise, conditional on the trained forecast series, and not adjusted for comparing several candidates. Paired stationary bootstrap; mean block length 21; 2,000 replications. The full aggregate JSON includes 5- and 63-observation block sensitivities with 1,000 replications each. Negative values favor the candidate.

D / Complete Activation Summary

Each row uses the corrected common test period and same-date VIX. Values are averaged across units and lags 1-20, then pooled across rolling folds.

Model	Activation	n	Pearson	Spearman
Pure LSTM	candidate	3738	+0.116	+0.071
Pure LSTM	forget	3738	-0.310	-0.025
Pure LSTM	input	3738	+0.264	+0.104
Pure LSTM	output	3738	+0.139	+0.104
Pure GRU	candidate	3738	+0.185	+0.213
Pure GRU	reset	3738	+0.034	+0.015
Pure GRU	update	3738	+0.417	+0.378
Feature LSTM	candidate	3738	+0.077	+0.075
Feature LSTM	forget	3738	-0.118	+0.054
Feature LSTM	input	3738	+0.288	+0.080
Feature LSTM	output	3738	+0.118	+0.084
Feature GRU	candidate	3738	+0.205	+0.230
Feature GRU	reset	3738	+0.052	+0.049
Feature GRU	update	3738	+0.329	+0.274
Ratio LSTM	candidate	3738	+0.130	+0.097
Ratio LSTM	forget	3738	-0.361	-0.114
Ratio LSTM	input	3738	+0.182	+0.116
Ratio LSTM	output	3738	-0.213	-0.144
Ratio GRU	candidate	3738	+0.231	+0.228
Ratio GRU	reset	3738	+0.025	+0.021
Ratio GRU	update	3738	+0.232	+0.185

These are ex-post correlations without interval estimates or a causal interpretation. Candidate activations are not sigmoid gates. Raw gate outputs, forecasts, and training logs are retained locally for inspection.

E / Reproducibility and Evidence

Package	Recorded version
numpy	1.26.4
pandas	2.2.3
scipy	1.15.3
arch	7.2.0
tensorflow	2.19.0
keras	3.10.0

Rerun commands

Install requirements-research.txt in an isolated environment.

1. Run scripts/run_corrected_experiment.py with --raw, --out, and --prepare.
2. Run scripts/run_all_models.py with the same --out and a bounded --workers value.
3. Run scripts/evaluate_corrected_experiment.py with --out, --destination, and the optional VIX snapshot.
4. Build the PDFs with reports/build_corrected_report.py from the aggregate results JSON.

Artifact identification

The aggregate results record the raw snapshot SHA-256, relevant source-code hashes, configuration, model prediction hashes, and matched VIX snapshot hash. The public repository omits raw market data and prediction CSVs. Hashes identify the local evidence but do not replace access to it for an independent full rerun.

Limits carried into the conclusions

One primary seed; one equity index; a cached historical dataset already examined in earlier research; a noisy daily proxy; log-objective mismatch; pointwise bootstrap uncertainty; no economic strategy or deployment test. The calendar slices and benchmark checks do not remove these limitations.

The underlying project originated in collaborative research. This revised portfolio report is presented by Eddie Liu. Contributor names and course identifiers are omitted; the byline does not imply sole authorship of the original experiments.

Aggregate evidence: reports/corrected_results.json. Source workflow and interpretation boundaries: docs/corrected-experiment-plan.md and docs/validation-status.md.